

Reflection on Code

Credit Name: CSE 3010 - Computer Science 3

Assignment Name: BreakAPlate.java

The following questions can help you in thinking critically about your problem-solving processes:

Understanding the Problem

How did you approach understanding the challenge?

Were there any parts of the problem you found confusing at first? If so, how did you resolve that confusion?

I approached the challenge by first understanding what the program was supposed to do when the Play button was clicked. I needed to make the program randomly choose between three different outcomes and display the correct images. At first, I was a little confused about how to generate a random number, but I figured it out by using `Math.random()` and converting the result into an integer from 1 to 3.

Planning the Solution

Did you create a plan or break the problem into smaller steps before coding?

How did you decide on the tools, data structures, or algorithms to use?

I broke the problem into smaller steps. First, I created the GUI with the button and labels, then I added the images, and finally I added the random selection and outcomes. I decided to use if and else if statements because there were three possible results.

Implementation

Did you write the code in small pieces or attempt the entire solution at once?

How did you handle moments when you felt stuck or unsure of what to do next?

I built the program in smaller pieces instead of trying to write everything at once. I first made sure the window and button worked, then added the images and finally programmed the random outcomes. When I got stuck, I looked at examples from previous programs and tested different parts of my code to find what was causing the problem.

Overcoming Challenges

What part of the problem was the most difficult for you?

How did you handle moments when you felt stuck or unsure of what to do next?

The most difficult part was getting the random number and making sure each number displayed the correct plate and prize. I handled this by breaking the problem down and checking each possible outcome separately. Testing the program multiple times helped me make sure all three results could occur.

Learning

Was there anything you learned that you think will help you with future challenges?

I learned more about using `Math.random()` and connecting it to button events in a GUI. I also learned that breaking a larger programming problem into smaller parts makes it easier to understand, test, and fix. This will help me with future projects because I can focus on one section of the program at a time.